

QUANTMASTER

Complete Product & Business Plan

17 modules · 500+ features · 128 mathematical simulations Dual-entity: SaaS + Hedge Fund
· AI Agent with live performance monitoring The most complete quantitative trading platform
ever designed

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1. EXECUTIVE SUMMARY

The Vision

QuantMaster is the platform that finally unifies everything a quantitative trader needs into a single, beautiful, professional-grade experience. Today, a serious quant trader must maintain accounts on TradingView (charts), QuantConnect (backtesting), Jupyter (data science), Bloomberg (data, \$2,400/month), Slack (team), GitHub (code) and still build their own live trading infrastructure. That is five to seven different tools, five invoices, five login credentials, five mental contexts to switch between, and zero integration between them. QuantMaster replaces all of them with one platform at a fraction of the cost.

The product is designed in three distinct layers. The first layer is what you see: TradingView-quality WebGL charts, a Monaco-based IDE (the same engine as VS Code), and a Zapier-style drag-and-drop visual builder for those who do not want to code. The second layer is what you analyze: 100+ mathematical models spanning options pricing, risk, portfolio optimization, time-series econometrics, and machine learning — plus 128 interactive simulations with Monte Carlo, stress testing, regime detection, and more. The third layer is what makes QuantMaster irreplaceable: an AI agent powered by Claude that writes code, validates it, runs backtests and, critically, monitors the strategy in live trading and tells you exactly why it diverges from its backtest and what to fix.

Layered beneath all of this is the dual-entity business model. QuantMaster Inc. is the SaaS company. QuantMaster Capital LP is a separate, independent hedge fund that uses anonymized aggregate patterns from the platform's 600,000+ users as research input. Two legal entities, two separate investor bases, no conflict of interest — but a self-reinforcing flywheel: more SaaS users generate better data signals, better signals generate higher hedge fund returns, higher returns attract elite traders who improve the SaaS product, which attracts more users.

Entity	Year 1	Year 2	Year 3	Year 5
SaaS ARR	\$1.9M	\$14M	\$56M+	\$400M+
Hedge Fund Net Profit	\$2.6M	\$15.8M	\$89.5M	\$359M
Combined Total	-\$12.9M (burn)	-\$5.95M (near-even)	\$107.6M profit	\$559M profit
SaaS Users	50K	200K	600K	2M+
Hedge Fund AUM	\$20M	\$100M	\$500M	\$2B

2. THE PROBLEM & MARKET OPPORTUNITY

The Fragmentation Problem

The quantitative trading software market is a paradox. It is used by some of the most technically sophisticated people on the planet — physicists, mathematicians, computer scientists — yet the tooling they are forced to use is fragmented, expensive, and built with zero consideration for user experience. A mid-tier hedge fund today pays \$2,400/month per terminal for Bloomberg, \$200/month for QuantConnect, builds its own Jupyter infrastructure, maintains separate GitHub repos, and still cannot get from an idea to a live strategy in a single workflow. The friction is enormous.

The retail trader has it worse. TradingView gives beautiful charts but its backtester is a toy: no realistic slippage, no walk-forward validation, no detection of overfitting. A retail trader backtests a strategy that shows a 300% return with a 2.0 Sharpe, deploys it live, and watches it bleed. Not because the trader is incompetent — but because the tool never told them that their strategy's Deflated Sharpe (adjusted for multiple testing) was 0.4, not 2.0. This is the silence that costs traders money. QuantMaster breaks it.

Key insight: The problem is not that the tools are bad in isolation. TradingView's charts are excellent. Bloomberg's data is excellent. The problem is that they were never designed to work together — and none of them have an AI layer that understands the full workflow.

Market Size

The addressable market is larger than it appears from the outside. The immediate market is the \$2.5 billion trading software and analytics market, growing at 15% annually and projected to exceed \$5 billion by 2030. The broader opportunity includes data services, API licensing, educational content, and strategy marketplace commissions — which collectively expand the total addressable market to \$8-10 billion. Within this, three distinct customer segments exist, each with different willingness to pay and different retention characteristics.

Segment	Size (Global)	WTP/Month	Current Tools	Pain Level
Retail Active Traders	500K+	\$50–\$200	TradingView, Yahoo	High (no real backtest)
Professional / Prop Traders	50K	\$200–\$500	QuantConnect, FactSet	High (no integration)
Hedge Funds (\$100M–\$5B AUM)	10K+	\$5K–\$15K	Bloomberg, custom	Extreme (\$2,400/seat)

3. THE DUAL-ENTITY BUSINESS MODEL

Why Two Separate Entities?

The dual-entity model is the architectural decision that transforms QuantMaster from a well-executed SaaS into a generational business. The premise is simple: the most valuable output of a platform used by 600,000 quantitative traders is not the subscription revenue — it is the anonymized, aggregate knowledge of what works and what does not in the market at any given moment. This knowledge, properly extracted and applied, is worth hundreds of millions of dollars annually as hedge fund alpha. Two separate legal entities allow this value to be captured without creating any conflict of interest with SaaS users.

The SaaS company — QuantMaster Inc. (C-Corporation, Delaware) — is a standard venture-backed software business. It takes subscription revenue, builds product, hires engineers, and grows users. The hedge fund — QuantMaster Capital LP (Limited Partnership) — is a completely separate legal entity with its own GP, its own LPs (family offices, HNW individuals), its own auditor, its own compliance officer, and its own legal counsel. The two entities share a brand and a philosophical approach, but they are operationally and legally independent. No SaaS investor owns hedge fund equity. No hedge fund LP owns SaaS equity. No user data is shared — only anonymized, aggregated statistical patterns that any quantitative research firm could theoretically derive from public market data.

	QuantMaster Inc. (SaaS)	QuantMaster Capital LP (Hedge Fund)
Legal Structure	C-Corporation (Delaware)	Limited Partnership
Revenue Model	Monthly/annual subscriptions	2% mgmt fee + 20% performance
Investors	Venture capital (Series A, B)	Family offices, HNW individuals
Year 3 Revenue	\$56M+ ARR	\$89.5M net profit
Valuation (Year 3)	\$2.4B–\$3B (20-30x ARR)	Keep private (perpetual income)
Exit	Sell or IPO	Keep forever
Team Lead	CEO / VP Product	Chief Investment Officer

The data flow between entities is governed by the SaaS Terms of Service, which explicitly state that anonymized, aggregated strategy patterns may be used for platform improvement, research, and analytics. No individual user can be identified from the aggregated data (GDPR-grade anonymization). The hedge fund treats this data as one input among many, alongside standard market data, academic research, and proprietary signal generation. Ultra Secure tier clients (\$50K+/month) receive a contractual guarantee of complete data isolation — their strategies are never analyzed, never aggregated, never used for any purpose outside their own account.

4. THE PLATFORM: MODULES 1-6

MODULE 1 — CHARTS & TERMINAL

The chart terminal is the first thing a user sees and the feature they will use every single day. It must be world-class or the product fails on first impression. QuantMaster uses TradingView's Lightweight Charts library — an open-source, Apache 2.0-licensed library with 25,000+ GitHub stars, maintained by TradingView Inc. itself. This library provides WebGL-accelerated rendering at 60 frames per second, capable of displaying one million candles without any visible lag. On this foundation, QuantMaster builds a feature layer that matches — and in several dimensions exceeds — TradingView's paid terminal.

Indicators are calculated server-side in Python (NumPy/SciPy), ensuring that the exact same calculation used in the visual terminal is used in the backtester. This eliminates one of the most common sources of backtest-to-live divergence: indicator discrepancy. When you add RSI(14) to your chart, the RSI(14) in your backtest is literally the same function call. No rounding differences, no vendor-specific calculation quirks.

• WebGL 60 FPS — 1M+ candles without lag	• Multi-pane synchronized charts (2x2 layout)
• 100+ technical indicators (server-side Python)	• Volume profile & market delta heatmaps
• Full drawing toolkit (Fibonacci, Elliott Wave, channels)	• Replay mode (rewind tick-by-tick)
• Multi-timeframe: 1s to 1M in same view	• Dark/light themes, custom color palettes
• Trade annotation overlay (from backtests)	• Screenshot & share chart state via URL

MODULE 2 — VISUAL STRATEGY BUILDER

The visual builder solves a specific user problem: most traders have clear intuitions about what should work, but cannot translate those intuitions into code. The builder uses a node-based canvas (similar to Zapier or n8n) where users drag blocks representing data sources, conditions, signals, and actions — then connect them with arrows to define the strategy flow. When satisfied, one click converts the entire canvas to valid, readable Python code. This is not a toy feature. The generated code uses the full QuantMaster SDK, handles position sizing, includes risk management blocks, and can be further edited in the code editor.

• Drag-and-drop canvas with 50+ block templates	• Auto-conversion to production Python code
• Conditional logic: if/then/else, AND/OR	• Position sizing blocks (fixed, Kelly, ATR-based)
• Backtests visible in same view as canvas	• 50 pre-built strategies ready to customize

MODULE 3 — CODE EDITOR (IDE)

For users who want full control, QuantMaster provides a professional-grade IDE built on Monaco — the same engine that powers Visual Studio Code. This means professional syntax highlighting, multi-cursor editing, code folding, and every keyboard shortcut developers already know. On top of Monaco, QuantMaster adds QuantMaster SDK autocomplete: as you type, it suggests valid functions, shows parameter documentation, and warns of common mistakes in real time. Jupyter-style cells allow running individual blocks of code and seeing results inline. Git integration means every version of every strategy is tracked, and the Sharpe ratio for each commit is stored alongside the code

diff.

• Monaco IDE (VS Code engine) — full feature parity	• Jupyter-style executable cells
• QuantMaster SDK autocomplete + inline docs	• Git: commits, diffs, branches, PRs
• AI inline assist (describe in comment, get code)	• Multi-file project structure support

MODULE 4 — BACKTESTER (INSTITUTIONAL GRADE)

The backtester is the technical heart of QuantMaster and the feature that most differentiates it from every competitor. It is event-driven (not vectorized), meaning it processes each market event in sequence and accurately simulates the timing constraints of real trading — you cannot buy at the open if your signal only fired after the close. The execution engine is written in Rust, compiled to a Python extension, which allows 20 years of daily data to be processed in under two seconds.

More important than speed is honesty. The QuantMaster backtester is the only one in any SaaS product that automatically computes and prominently displays the Deflated Sharpe Ratio (Bailey & Lopez de Prado, 2014). The Deflated Sharpe accounts for the number of strategy variations tested, the length of the backtest, skewness, and kurtosis. A strategy with a raw Sharpe of 2.0 that was discovered after testing 50 variations on 5 years of data might have a Deflated Sharpe of 0.3 — meaning it is almost certainly the product of data mining, not a real edge. QuantMaster shows this number prominently, with an explanation in plain English. No competitor does this. It is the feature that builds institutional trust.

• Event-driven (no look-ahead by design)	• Rust core: 20 years in under 2 seconds
• Almgren-Chriss slippage model (realistic)	• Borrow costs, financing, dividends included
• Deflated Sharpe — automatic overfitting detection	• Probabilistic Sharpe with confidence interval
• Walk-forward validation (anchored & rolling)	• Monte Carlo: 1,000 simulation paths
• Benchmark comparison (SPY, custom)	• Trade blotter with every entry/exit annotated
• Monthly returns heatmap	• Stress testing (2008, COVID, custom scenarios)

MODULE 5 — 100+ MATHEMATICAL MODELS

The mathematical models library is QuantMaster's answer to Bloomberg's BQNT and FactSet's analytics suite — but accessible to anyone at \$99/month. Each model is a self-contained app: a card with parameter inputs, a results panel with charts and statistics, and an 'Apply to Strategy' button that integrates the output directly into your backtest. The library spans options pricing (Black-Scholes through Heston and SABR), fixed income (bond pricing, duration, convexity, yield curve bootstrapping), portfolio optimization (Markowitz through Hierarchical Risk Parity), risk (VaR, CVaR, copulas, stress testing), volatility (GARCH through DCC), and machine learning models with walk-forward validation built in.

MODULE 6 — DATA INTEGRATION

Data is the fuel of any quantitative strategy. QuantMaster provides a tiered data architecture that begins completely free and scales to Bloomberg-grade institutional data as the user's needs grow. The Free tier uses Yahoo Finance for equities, CoinGecko for crypto, and the Federal Reserve's FRED API for macroeconomic data — all completely free, no API key required from the user. The Pro tier adds Polygon.io, which provides 20+ years of clean, adjusted historical

data with minute-level resolution, real-time quotes under one second, and options chain data including implied volatility and Greeks. Enterprise tier adds Bloomberg Terminal API and Reuters/Refinitiv, bringing the full institutional data stack into the same interface.

Tier	Data Provider	History	Real-Time	Options	Alt Data	Cost
Free	Yahoo + FRED + CoinGecko	5 years	15 min delay	No	No	\$0
Pro	+ Polygon.io	20+ years	<1 second	Yes	No	\$500/mo
Pro+	+ Level 2 Quotes	20+ years	<1 second	Full chain	No	\$700/mo
Enterprise	+ Bloomberg/Reuters	40+ years	<100ms	Institutional	Yes	Included

5. MODULE 7 — THE AI AGENT

The Agent That Changed Everything

The AI agent is the product's most important differentiator. Not because AI is fashionable, but because quantitative trading has a specific unsolved problem that AI is uniquely positioned to solve: the gap between a trader's intuition and a working, validated, production-ready strategy. Currently, this gap takes days or weeks to bridge — even for experienced quants. The QuantMaster AI agent closes it in minutes.

The agent is powered by Claude, which is particularly strong at both code generation and structured reasoning. In the quantitative finance domain, this translates into code that is not just syntactically correct but financially sound: proper handling of look-ahead bias, correct calculation of returns, appropriate position sizing, and realistic cost assumptions. The agent knows the QuantMaster SDK in detail and generates code that immediately runs in the IDE without modification in the vast majority of cases.

Core Generation Capabilities

A user can describe a strategy in plain English — 'Buy when the 10-day RSI crosses above 30 on stocks in the S&P; 500 with momentum above their 200-day moving average, size positions using ATR, limit each position to 5% of portfolio' — and the agent generates a complete, runnable Python strategy in under ten seconds. It then automatically triggers a backtest, displays the results in the split-screen workspace, highlights any concerns (look-ahead bias, low Deflated Sharpe, insufficient sample size), and suggests specific improvements. The conversation is multi-turn: the user can refine the strategy iteratively across dozens of exchanges without losing context.

• English → Python: full strategy in under 10 seconds	• Automatic backtest on generation
• Look-ahead bias scanner (flags before running)	• Overfitting check (Deflated Sharpe displayed)
• Multi-turn: refine across entire session	• Context memory: past strategies, regime, risk limits
• Explains every code decision in plain English	• Suggests improvements with evidence from backtest

NEW CAPABILITY: Live Performance Monitoring & Adaptive Refinement

This is the feature that makes QuantMaster irreplaceable. The single biggest reason traders abandon good strategies is not that the strategies are bad — it is that performance diverges from backtest expectations in live trading, the trader does not understand why, and panics. The QuantMaster AI agent solves this by monitoring live and paper trading performance continuously and diagnosing divergences automatically.

From the moment a strategy is deployed to live or paper trading, the agent begins tracking a set of performance metrics and comparing them to the expected ranges derived from the backtest. If the live Sharpe ratio falls more than 20% below the backtest Sharpe, the agent fires an alert. But it does not just alert — it diagnoses. It analyzes the current market regime (using HMM), checks whether volatility has changed (GARCH), tests whether correlation structures have broken (DCC-GARCH), and checks whether the strategy's signals are still firing as expected. Within minutes, it presents a root cause analysis in plain English with specific, testable fixes — each of which it can immediately backtest on the same historical data to show whether the fix helps.

Example — Week 4 of live trading: Your momentum strategy shows a win rate of 52% (backtest projected 64%). The agent diagnoses: 'Market entered a consolidation regime on Tuesday (HMM confidence: 87%). Momentum strategies historically return -0.8% per month in consolidation. Your current momentum lookback (20 days) is too short for this regime. I tested a 45-day lookback with an ADX filter (>25): Sharpe improves from 0.4 to 1.7 in consolidation periods. Want me to deploy this version to paper trading?'

The Adaptive Loop in Practice

The monitoring and refinement system operates on a continuous feedback loop that has five phases. In phase one (deployment), the agent creates a performance model from the backtest results — expected Sharpe range, projected win rate, maximum expected drawdown, and typical trade duration. In phase two (monitoring), the agent compares incoming live data against these expectations, updating statistical significance estimates daily. In phase three (detection), when divergence exceeds a threshold and reaches statistical significance (typically 15-20 trading days), the agent activates. In phase four (diagnosis), it runs a systematic root cause analysis. In phase five (refinement), it presents tested fixes with before/after performance comparisons.

Divergence Type	What the Agent Detects	What It Suggests
Regime change	HMM detects state shift (bull→bear→choppy)	Add regime filter, adjust lookback
Vol spike	GARCH shows vol 2x+ higher than backtest	Scale positions down, add vol filter
Correlation break	DCC shows pairs correlation collapsed	Pause pairs trade, recalibrate
Market impact	Real slippage 3x Almgren-Chriss prediction	Smaller size, limit order execution
Signal decay	Win rate declining over rolling 20 days	Add freshness filter, reduce universe
Drawdown pace	Drawdown speed 2x faster than projected	Tighten stop loss, reduce leverage

Proactive Morning Intelligence Brief

Every morning before market open, the agent sends a brief to each active user summarizing the current market regime (with confidence level), how each live strategy is performing relative to expectations, any signals of emerging divergence, and the day's highest-probability opportunities based on the user's historical strategy preferences. This is not a generic newsletter — it is personalized to each user's specific portfolio, strategies, and risk tolerance.

6. MODULES 8-16: THE FULL ECOSYSTEM

MODULE 8 — LIVE TRADING & EXECUTION

The live trading module connects QuantMaster to real brokerage infrastructure, transforming backtested strategies into live positions with a single click. Supported brokers at launch include Alpaca (US equities and crypto, commission-free), Interactive Brokers (global equities, options, futures, forex), Binance and Coinbase (crypto). The module supports ten order types: market, limit, stop, stop-limit, trailing stop, VWAP, TWAP, iceberg, bracket, and OCO. Pre-trade risk controls include maximum position size, maximum daily loss, maximum portfolio concentration, and sector exposure limits. A kill switch (PIN-protected) halts all live trading instantly. Paper trading is identical to live trading in every respect except that orders are simulated rather than sent to a broker — the same interface, the same fills (using real bid/ask data), the same risk controls.

• Alpaca, IB, Binance, Coinbase integration	• Market, limit, stop, bracket, VWAP, TWAP, iceberg
• Pre-trade risk controls (position size, daily loss limit)	• Kill switch (PIN-protected)
• Paper trading (identical UX, real bid/ask simulation)	• Real-time P&L; and risk metrics

MODULE 9 — GITHUB INTEGRATION

Version control is non-negotiable in professional software development. It should be equally non-negotiable in quantitative strategy development, yet no trading platform has ever implemented it properly — until QuantMaster. The GitHub integration is bidirectional: strategies created in QuantMaster can be pushed to a connected repo, and strategies in a repo can be pulled into QuantMaster. Every commit to a strategy automatically runs a backtest and stores the resulting metrics (Sharpe, CAGR, MaxDD) in the commit message. Diffs between strategy versions show both code changes and performance changes side by side. Pull requests include a backtest comparison between branches. Team lead approval is required before any strategy goes live.

• Bidirectional GitHub sync	• Sharpe/CAGR/MaxDD stored per commit
• Code diff + performance diff side by side	• PR workflow with backtest comparison
• Branch strategies like software (feature/fix branches)	• Team lead approval for live deployment

MODULE 10 — STRATEGY MARKETPLACE

The marketplace creates a new economic model for quantitative strategy development: traders who develop good strategies can license them to others and earn recurring income without revealing the source code. A strategy listed on the marketplace shows its live track record (verified by QuantMaster — not self-reported), the gap between backtest and live performance (a key trust signal), the Deflated Sharpe, maximum drawdown, and a 90-day equity curve preview. Buyers subscribe to a strategy for a monthly fee; the developer earns 70% of that fee (QuantMaster takes 30%). Strategies with live track records longer than six months and a gap below 15% earn a 'Verified' badge.

• Live track record (verified, not self-reported)	• Backtest-to-live gap displayed prominently
• 70/30 revenue split (creator/platform)	• Strategy subscriptions (monthly recurring)
• 'Verified' badge: 6+ months live, gap <15%	• Searchable by asset class, Sharpe, drawdown

MODULE 11 — PORTFOLIO ANALYTICS

The portfolio analytics module aggregates all live and paper strategies into a single consolidated view. The correlation matrix (updated daily using DCC-GARCH) shows how each strategy correlates with others, making concentration risk visible. Factor attribution decomposes portfolio returns into contributions from market, size, value, momentum, and quality factors — showing how much alpha is truly factor-independent. Consolidated VaR combines all positions across brokers and strategies into a single risk estimate. Stress testing applies historical and hypothetical scenarios to the full portfolio simultaneously.

• Consolidated P&L; across all strategies and brokers	• Correlation matrix (DCC-GARCH, daily)
• Factor attribution (Fama-French 5-factor)	• Consolidated VaR/CVaR across portfolio
• Historical + hypothetical stress tests	• Portfolio optimization (rebalancing suggestions)

MODULE 12 — ENTERPRISE & COMPLIANCE

Enterprise clients have compliance requirements that consumer SaaS products typically ignore. QuantMaster's enterprise module was designed from the ground up for regulated environments. The immutable audit trail logs every action — who clicked what, when, with what result — in a tamper-proof, PDF-exportable format. Six role types (Viewer, Analyst, Trader, Risk Officer, Compliance Officer, Administrator) allow granular access control. Single Sign-On via SAML/OAuth integrates with the client's existing identity provider. SOC 2 Type II certification is planned for Month 6 of Year 2. The Ultra Secure tier provides complete infrastructure isolation: a dedicated compute environment, BYOK (bring your own encryption keys), and a contractual guarantee of zero data analysis.

• Immutable audit trail (every action logged, PDF-exportable)	• Six role types: Viewer, Analyst, Trader, Risk Officer, Compliance, Admin
• SSO via SAML/OAuth (integrates with client's identity provider)	• SOC 2 Type II (Year 2)
• Ultra Secure: dedicated infra + BYOK + contractual data isolation	• Regulatory report generation (pre-formatted)

MODULES 13-16 — Intelligence, Alerts, Education, API

Module 13 (Market Intelligence) aggregates anonymized behavioral data from all platform users to surface collective intelligence: what percentage of users are currently long momentum, where conviction is building, what regime the platform's aggregate signal detector has identified. No other tool has this because no other tool has 600,000 active quantitative traders. Module 14 (Alerts) supports ten alert types delivered across eight channels including Slack, Discord, Telegram, SMS, email, and webhook — all configurable with custom thresholds and logic. Module 15 (Education) provides 50+ example strategies with full explanations, structured courses on every mathematical model, and certifications that users can display on LinkedIn. Module 16 (API) exposes the full platform programmatically: REST API, webhooks, Python SDK, and Zapier integration for no-code automation.

7. MODULE 17 — 128 MATHEMATICAL SIMULATIONS

The Most Complete Simulation Library in Any Trading SaaS

Module 17 is the feature that no competitor can replicate quickly. Building a simulation library is not difficult in principle — the mathematics is documented in academic literature. The difficulty is in implementation quality: each simulation must be numerically stable, fast enough to run interactively, calibrated correctly to real market data, and integrated with the rest of the platform so that outputs flow directly into backtests and the AI agent. QuantMaster's 128 simulations span seventeen categories, from simple Black-Scholes option pricing to advanced Bayesian filtering and Riemannian geometry. They are organized in an App Store-style interface: each simulation is a card, each card has parameter inputs, a live results panel, and an 'Apply' button. Users pay in Simulation Credits (bundled with each tier) based on computational complexity.

CATEGORY 1: Price Path Models (8 Simulations)

Price path models are the foundation of quantitative finance: mathematical descriptions of how asset prices evolve through time. The baseline is Geometric Brownian Motion (GBM), which underlies the original Black-Scholes model and assumes continuously compounded returns are normally distributed. While GBM is widely used, its limitations are well-known: it cannot reproduce fat tails, volatility clustering, or discontinuous jumps. QuantMaster includes progressively more realistic alternatives. Merton's Jump Diffusion adds a Poisson process on top of GBM to model sudden price gaps (earnings surprises, geopolitical events). The Heston model makes volatility itself a random process, which reproduces the volatility smile observed in options markets. The Lévy process generalizes jump diffusion to allow jumps of arbitrary size and frequency, producing significantly fatter tails. Fractional Brownian Motion introduces a Hurst exponent that captures long-range dependence — the fact that markets exhibit autocorrelation over much longer horizons than classical theory predicts.

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| • Geometric Brownian Motion (GBM) — classical baseline, foundation of Black-Scholes |
| • Jump Diffusion (Merton) — Poisson jumps capture earnings gaps and event-driven moves |
| • Heston Stochastic Volatility — volatility is a random process (more realistic) |
| • Lévy Process — fat tails, arbitrary jump sizes, better tail risk modeling |
| • Fractional Brownian Motion — Hurst exponent captures long-range dependence in returns |
| • Ornstein-Uhlenbeck Process — mean reversion, used for pairs trading and spreads |
| • Vasicek-style General Mean Reversion — applicable to any asset, not just rates |
| • Regime-Switching Brownian Motion — different drift/vol parameters per market state |

CATEGORY 2: Volatility Models (8 Simulations)

Volatility models are essential for any strategy involving options, risk management, or dynamic position sizing. The GARCH family models the well-documented phenomenon of volatility clustering — large moves tend to follow large moves. GARCH(1,1) is the industry standard: simple, fast, and sufficient for most applications. EGARCH extends this to capture the leverage effect (negative returns increase volatility more than positive returns of the same magnitude), a widely observed asymmetry in equity markets. For multi-asset portfolios, DCC-GARCH (Dynamic Conditional Correlation) is essential: it models how correlations between assets change over time, especially under stress when correlations tend to spike toward 1.0 simultaneously.

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| • GARCH(1,1) — volatility clustering, industry standard baseline |
| • EGARCH — asymmetric (bad news increases vol more than good news) |
| • GJR-GARCH — asymmetric term specifically for negative return shocks |
| • Markov-Switching GARCH — different vol parameters in different market regimes |
| • DCC-GARCH — dynamic correlation matrix (critical for multi-asset portfolios) |
| • Multivariate GARCH (BEKK) — full time-varying covariance matrix |
| • Volatility Term Structure — how implied vol changes at different maturities |
| • Jump Intensity Models — Hawkes processes for volatility clustering and self-excitation |

CATEGORY 3: Option Pricing (14 Simulations)

The options pricing library covers the full spectrum from the classical Black-Scholes formula to state-of-the-art stochastic volatility models and numerical methods for exotic options with no closed-form solution. Black-Scholes remains the lingua franca of options pricing despite its unrealistic assumptions, primarily because every practitioner understands it and its Greeks have intuitive interpretations. For more accurate pricing, the Heston model is preferred: its stochastic volatility process reproduces the volatility smile without requiring local volatility surfaces. The SABR model is particularly popular in fixed income and FX markets because it provides closed-form approximations for implied volatility at any strike, making calibration fast and practical.

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| • Black-Scholes (European) — closed-form, the universal industry baseline |
| • Binomial Tree — American options with optimal early exercise boundary |
| • Trinomial Tree — better lattice convergence than binomial for same step count |
| • Monte Carlo for Exotics — path-dependent payoffs: barrier, Asian, lookback |
| • Heston Model — stochastic vol, reproduces volatility smile accurately |
| • Jump Diffusion (Bates) — combines Heston with Merton jumps for equity options |
| • SABR Model — closed-form vol approximation, popular in FX and rates |
| • Local Volatility (Dupire) — calibrates to entire market vol surface |
| • Stochastic Local Volatility (SLV) — combines Heston + Local Vol |
| • Barrier Options — knock-in, knock-out with single and double barriers |
| • Lookback Options — floating and fixed strike, max/min dependent payoff |
| • Asian Options — arithmetic and geometric averaging |
| • Chooser Options — right to choose call or put at a future date |
| • Rainbow / Spread Options — multi-asset, outperformance, exchange options |

CATEGORY 4: Greeks & Dynamic Hedging (10 Simulations)

Understanding option sensitivity is not just theoretical — it is the foundation of practical risk management for any portfolio containing non-linear instruments. The Greeks quantify how an option's value changes with respect to each market variable. Delta is the first-order price sensitivity; Gamma is the rate of change of Delta (crucial for understanding hedging costs); Vega measures sensitivity to implied volatility (the dominant risk for long option positions); Theta measures time decay. QuantMaster computes all Greeks analytically where formulas exist and numerically via Monte Carlo for exotic options. The Delta Hedging simulation is particularly educational: it shows, day by day, how much a delta-neutral hedge costs to maintain, and how Gamma Scalping (being long gamma while delta-neutral) profits from realized volatility exceeding implied volatility.

• Delta — first derivative of option price w.r.t. underlying
• Gamma — rate of delta change (hedging rebalancing cost indicator)
• Vega — sensitivity to implied volatility changes
• Theta — daily time decay (P&L; erosion for long option positions)
• Rho — sensitivity to interest rate changes (matters for long-dated options)
• Vanna — cross-derivative of delta w.r.t. vol (key for vol surface trading)
• Volga — second derivative w.r.t. vol (curvature of the vol smile)
• Delta Hedging Simulation — daily rebalancing costs over full strategy lifetime
• Gamma Scalping — long gamma, short delta: profits from high realized vol
• Portfolio Greeks Aggregation — real-time aggregate sensitivities across all positions

CATEGORY 5: Risk Models (10 Simulations)

Risk measurement is one area where the gap between retail tools and institutional practice is most stark. Most retail platforms compute VaR using a single method (usually parametric, assuming normality) and display it without any uncertainty estimate. QuantMaster implements four VaR methods and shows their results side by side, making the model risk visible. Cornish-Fisher VaR adjusts the normal quantile for the observed skewness and excess kurtosis of the return distribution — for assets with fat tails, this can make VaR 30-50% larger than the naive estimate. CVaR (Conditional VaR, also known as Expected Shortfall) is the average loss in the worst 1% of scenarios — a more coherent risk measure that is increasingly preferred by regulators and risk managers.

• VaR Historical — empirical percentile, non-parametric, robust to non-normality
• VaR Parametric — normal distribution assumption, fast but optimistic in tails
• VaR Monte Carlo — flexible, handles any return distribution
• VaR Cornish-Fisher — adjusts for skewness and kurtosis (critical for equity and crypto)
• CVaR / Expected Shortfall — average of losses beyond VaR (coherent risk measure)
• Drawdown Analysis — underwater plot, time to recovery, maximum drawdown statistics
• Copula Models (Gaussian, Clayton, Student-t) — models tail dependence between assets
• Ledoit-Wolf Shrinkage — robust covariance estimation (reduces estimation noise)
• Factor Risk Decomposition — attributes P&L; to market, size, value, momentum
• Tail Risk Parity — weights assets by tail risk contribution, not volatility

CATEGORY 6: Portfolio Optimization (9 Simulations)

Portfolio optimization theory dates to Markowitz's 1952 paper but the practical implementation is far from trivial. Classic mean-variance optimization is notoriously sensitive to input assumptions: small changes in expected return estimates produce dramatically different portfolio weights, often resulting in corner solutions (one asset gets nearly all the weight). QuantMaster implements robust alternatives. Black-Litterman combines the market equilibrium returns (from CAPM) with the investor's specific views, with a parameter (τ) controlling how much weight to give each — producing portfolios that are more stable and intuitive. Hierarchical Risk Parity (HRP), proposed by Lopez de Prado, uses hierarchical clustering to group assets and then applies risk-parity weighting within each cluster — it has been shown to outperform Markowitz out-of-sample in numerous studies.

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| • Mean-Variance (Markowitz) — efficient frontier with any objective function |
| • Minimum Variance — lowest possible portfolio volatility |
| • Maximum Sharpe — best risk-adjusted return (tangency portfolio) |
| • Black-Litterman — combines CAPM equilibrium with investor views |
| • Risk Parity — each asset contributes equally to total portfolio risk |
| • Minimum CVaR — minimizes worst-case average loss (more robust than min-variance) |
| • Hierarchical Risk Parity (HRP) — clustering-based, more robust out-of-sample |
| • Fama-French Factor Optimization — target factor exposures explicitly |
| • Rebalancing Optimizer — balance return improvement against transaction costs |

CATEGORY 7: Time Series Econometrics (8 Simulations)

Time series econometrics forms the quantitative backbone of many systematic strategies, particularly those involving cointegration (pairs trading), mean reversion, and predictive models. The Johansen cointegration test is the standard method for identifying pairs of assets whose prices move together in the long run — even if both individual price series are non-stationary (random walks). Finding a cointegrated pair means finding a stationary spread that will mean-revert to its historical equilibrium, which is the foundation of statistical arbitrage. VECM (Vector Error Correction Model) takes this further by modeling both the short-term dynamics and the long-run equilibrium relationship simultaneously.

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| • ARIMA/SARIMA — auto-selected order via AIC/BIC criterion, seasonal components |
| • VAR (Vector Autoregressive) — interdependence across multiple time series |
| • VECM (Vector Error Correction) — cointegrated series with equilibrium dynamics |
| • Johansen Cointegration Test — identify statistically significant pairs |
| • Granger Causality — test whether X has predictive power over Y |
| • ADF/KPSS Unit Root Tests — determine if a series is stationary |
| • Phillips-Perron Test — robust unit root test (handles serial correlation differently) |
| • Spectral Analysis — frequency domain analysis (seasonality, cycles) |

CATEGORY 8: Machine Learning Models (9 Simulations)

Machine learning in quantitative finance must be approached with extraordinary caution. The primary failure mode is overfitting: a model that explains past data perfectly but has zero predictive power going forward. QuantMaster's ML simulation suite is built around this risk. Every ML model includes mandatory walk-forward validation (training only on past data at each step), feature importance with SHAP explanations (so the user understands what the model is actually using), and out-of-sample Sharpe estimation with confidence intervals. The Reinforcement Learning module is particularly experimental — it allows users to define a trading environment (state space, action space, reward function) and train agents using DQN or A3C — but includes strong warnings about the data requirements needed to avoid overfitting in high-dimensional action spaces.

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| • Linear Models — OLS, Ridge, Lasso, ElasticNet with cross-validation |
| • Random Forest — feature importance, SHAP values, out-of-bag error |
| • Gradient Boosting — XGBoost, LightGBM, CatBoost with GPU acceleration |
| • LSTM — sequence model for temporal dependencies in price and volume data |
| • GRU — lighter alternative to LSTM, fewer parameters |
| • Attention / Transformer — self-attention mechanism for longer-range dependencies |
| • Ensemble Methods — stacking, voting, blending for improved generalization |
| • Walk-Forward Validation — anchored and rolling, mandatory for all ML models |
| • Reinforcement Learning — DQN, A3C for position sizing and execution optimization |

CATEGORY 9: Execution & Market Impact (8 Simulations)

Execution costs are the graveyard of backtests that looked great on paper. Almgren and Chriss (2001) developed the first rigorous mathematical framework for optimal trade execution: given a target quantity to trade, what schedule minimizes the combination of market impact cost and timing risk? Their model decomposes market impact into a temporary component (price reverts after your trade) and a permanent component (price incorporates your information signal permanently). QuantMaster implements this model fully, calibrated to historical liquidity data for each asset. Kyle's Lambda measures the informativeness of order flow — how quickly prices adjust when an informed trader acts — and is used to estimate adverse selection costs for strategies that rely on being fast (momentum, event-driven).

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| • Almgren-Chriss Optimal Execution — minimize cost + risk tradeoff |
| • Temporary Market Impact — price reverts after trade completes |
| • Permanent Market Impact — price incorporates information signal permanently |
| • Kyle's Lambda — adverse selection cost from informed order flow |
| • Volatility-Adjusted Impact — impact scales non-linearly with realized vol |
| • Seasonal Liquidity Patterns — impact differs by time of day, day of week |
| • Venue-Specific Impact — NYSE vs NASDAQ vs dark pool differences |
| • Order Book Depth Simulation — how your order moves through the book |

CATEGORY 10: Market Regimes (6 Simulations)

Market regimes are the single most important contextual variable for any trading strategy. A momentum strategy that returns 25% annually in bull markets may lose 15% annually in bear markets. A mean-reversion strategy that works beautifully in choppy markets will bleed slowly in trending markets. Yet most trading platforms provide no tools for regime identification. QuantMaster implements the Hidden Markov Model (HMM) as the primary regime detector. The HMM assumes the market switches between K hidden states (typically 3: bull, bear, and sideways/choppy) according to a transition probability matrix. At each point in time, it assigns a probability to each state based on the observed returns and volatility. This probability is displayed in real-time on the chart terminal, giving the trader a continuous read on market character.

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| • Hidden Markov Model (3 states) — bull/bear/choppy probability in real time |
| • Markov-Switching Regression — strategy betas change per regime |
| • Threshold Autoregressive (TAR) — regime defined by vol or momentum threshold |
| • Entropy-Based Detection — Shannon entropy of returns (high = chaotic) |
| • Regime-Dependent Factor Loading — different factor weights per state |
| • Regime Duration Analysis — statistical properties and transition probabilities |

CATEGORY 11: Stress Testing & Scenarios (8 Simulations)

Stress testing answers the question every risk manager is actually worried about: 'What happens to my portfolio in a scenario I have not planned for?' Historical scenarios are the most credible: they use actual price movements from the worst market dislocations in modern history. The 2008 Financial Crisis scenario reproduces the full timeline from August 2007 to March 2009, including the correlation collapse in September 2008 when virtually every asset class fell simultaneously. The 2020 COVID scenario reproduces the fastest 30%+ drawdown in market history (23 trading days), including the subsequent V-shaped recovery that caught many risk managers off guard. Reverse stress testing is conceptually different: instead of asking 'what does scenario X do to my portfolio,' it asks 'what scenario causes a 20% drawdown in my portfolio?' This is more actionable for contingency planning.

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| • 1987 Black Monday — 19.7% single-day decline, portfolio behavior |
| • 2008 Financial Crisis — full 18-month stress including correlation collapse |
| • 2020 COVID Crash — 35% in 23 days + V-shaped recovery |
| • 2010 Flash Crash — 5-minute 10% decline and recovery mechanics |
| • Fed Rate Shock (+200bp) — bond, equity, credit, and currency impact |
| • Recession Scenario — GDP -2%, earnings -30%, systematic repricing |
| • Stagflation — inflation 8%, growth 0%, worst case for traditional 60/40 |
| • Reverse Stress Test — find the scenario that causes your target loss amount |

CATEGORY 12: Credit & Fixed Income (6 Simulations)

Fixed income and credit instruments require specialized mathematical tools that are absent from virtually all retail-focused trading platforms. Bond pricing is conceptually simple — present value of future cash flows — but practical implementation requires handling day count conventions, yield curve bootstrapping, accrued interest, and embedded options. The Vasicek and CIR models describe the evolution of the short-term interest rate as a mean-reverting stochastic process, allowing the pricing of interest rate derivatives and the simulation of yield curve scenarios. CDS (Credit Default Swap) pricing requires modeling the hazard rate of the reference entity — the probability per unit time of default — and deriving the fair spread that equates the expected protection leg to the expected fee leg.

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| • Bond Pricing — fixed/zero/floating coupon, accrued interest, day count |
| • Yield-to-Maturity — Newton-Raphson solver, modified duration calculation |
| • Duration and Convexity — Macaulay, modified, dollar duration |
| • Yield Curve Bootstrapping — extract spot rates from coupon bond prices |
| • Vasicek / CIR Interest Rate Models — stochastic rate simulation |
| • CDS Spread Pricing — hazard rate model, protection and fee leg |

CATEGORY 13: Correlation & Copulas (7 Simulations)

Correlation is the most commonly misused concept in portfolio management. The standard Pearson correlation captures only linear dependence and assumes the relationship between two variables is constant — neither of which is true in financial markets. Correlation between assets tends to spike during market stress (when diversification is needed most) and collapse during calm periods. DCC-GARCH (Dynamic Conditional Correlation) models this time-variation explicitly. Copulas provide an even richer framework: by separating the marginal distributions (the behavior of each asset in isolation) from the dependence structure (how they move relative to each other), copulas can model situations where two assets are independent most of the time but become highly correlated in the tails — exactly what happens during market crises.

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| • Rolling Correlation — moving-window Pearson correlation |
| • Exponential Weighted Correlation — recent observations weighted more heavily |
| • DCC-GARCH — dynamic conditional correlation (time-varying, regime-aware) |
| • Principal Component Analysis (PCA) — find latent factors driving co-movement |
| • Gaussian Copula — models linear tail dependence (understates crisis correlation) |
| • Clayton / Student-t Copula — models heavy lower-tail dependence |
| • Tail Dependence Estimation — measure probability of joint extreme losses |

CATEGORY 14: Bayesian & Advanced Methods (4 Simulations)

Bayesian methods provide a coherent framework for updating beliefs as new data arrives — exactly what a trading system needs to do continuously. The Kalman Filter is the optimal Bayesian estimator for linear Gaussian systems: given a noisy observation of a latent state (say, the 'true' value of a spread), it produces the minimum-variance unbiased estimate. In pairs trading, the Kalman Filter provides a dynamic hedge ratio that adapts as the relationship between two assets evolves — far superior to a fixed OLS regression estimated on a historical window. The Particle Filter extends this to non-linear and non-Gaussian systems, at the cost of requiring Monte Carlo approximation.

• Kalman Filter — optimal state estimation for linear systems (pairs trading hedge ratio)
• Particle Filter — non-linear, non-Gaussian Bayesian filtering
• Bayesian Inference — prior + likelihood → posterior updating
• Hierarchical Bayesian Model — share information across related assets

Simulation Credits — The Pricing Model

Not all simulations are computationally equal. A Black-Scholes calculation takes 0.1 seconds on a CPU. Training an LSTM network can take two minutes on a GPU. Pricing 100,000 Monte Carlo paths for a portfolio of exotic options can take five minutes. A flat subscription that includes all simulations would either be priced so high that users feel gouged when running simple models, or priced so low that the infrastructure costs are unsustainable. The Simulation Credits model solves this elegantly: each user tier includes a monthly credit budget. Each simulation costs a number of credits proportional to its computational cost. Simple models are cheap or free within any tier. Complex models are accessible but consume more credits, naturally incentivizing users who need high-complexity work to move to higher tiers.

Simulation	Cost (Credits)	Est. Time	Available From
Black-Scholes (European option)	1	<0.1s	Free
GARCH(1,1)	2	0.5s	Free
VaR Historical	2	0.2s	Free
ARIMA (auto-select p,d,q)	5	1s	Free
Johansen Cointegration	8	2s	Free
Monte Carlo (1,000 paths)	20	5s	Free
HMM Regime Detection	50	10s	Free
Heston Model	50	10s	Pro
Monte Carlo (10,000 paths)	100	15s	Pro
DCC-GARCH	200	30s	Pro
Walk-Forward Validation (full)	300	45s	Pro
Monte Carlo (100,000 paths)	500	90s	Pro+
LSTM Training	1,000	120s	Pro+
Full stress test battery (all 8)	500	90s	Pro+

Tier	Credits/Month	Price	Effective Cost / Credit	Overflow Rate
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Free	50	\$0	N/A	No
Pro	2,000	\$99	\$0.0495	+\$0.10/credit
Pro+	10,000	\$199	\$0.0199	+\$0.05/credit
Enterprise	100,000+	\$5K–\$15K custom	\$0.05–0.15	Negotiated
Ultra Secure	Unlimited	\$50K+	Included	N/A (dedicated GPU)

8. QUANTMASTER CAPITAL — HEDGE FUND

The Thesis

QuantMaster Capital LP is predicated on a single insight: a platform used by 600,000 quantitative traders generates a form of crowd intelligence that has never before been available. Not the individual strategy of any one trader — that would be illegal and unethical — but the aggregate statistical signal: what percentage of the platform's users are currently positioned long versus short, where conviction is building before it appears in price action, and what the aggregate backtest quality distribution looks like for each asset class. This information, properly anonymized and aggregated, is a form of alternative data that Bloomberg does not sell and no other firm can generate without an equivalent platform.

The fund operates five core quantitative strategies, each running on separate capital sub-allocations with independent risk limits. The strategies were selected to be complementary rather than correlated: statistical arbitrage (mean-reversion) is negatively correlated with momentum; volatility surface trading is generally uncorrelated with both; fixed income relative value provides stability during equity stress. The CIO and risk officer are hired separately from the SaaS team and report to a separate board of directors.

Strategy	Allocation	Expected Return	Risk	Core Model
Statistical Arbitrage	30% / \$150M	12-18%	Low vol	LSTM + anomaly detection
Mean Reversion	20% / \$100M	10-15%	Medium vol	GARCH + OU process
Cross-Asset Momentum	25% / \$125M	15-20%	Higher vol	HMM regime filter
Vol Surface Arb	20% / \$100M	10-14%	Low vol	Heston mispricings
Pairs Trading	5% / \$25M	8-12%	Very low	Johansen + Kalman Filter

Year	AUM	Gross Return	Net Return	Management Fee	Performance Fee	Net to LP
1	\$20M	18%	15%	\$400K	\$600K	\$2.0M
2	\$100M	20%	18%	\$2M	\$3.2M	\$14.8M
3	\$500M	22%	20%	\$10M	\$20M	\$89.5M
5	\$2B	20%	18%	\$40M	\$64M	\$359M

9. PRICING & MONETIZATION

The Psychology of the Pricing Architecture

Every number in the pricing table was chosen deliberately. The Free tier exists not as a charity but as the primary user acquisition channel. Five backtests per month (not ten, not unlimited) is precisely calibrated to allow a new user to discover the platform's depth and hit the limit at the exact moment of maximum enthusiasm — when they have just found something promising and want to test more variations. At that moment, \$99/month feels trivial. The limit is not arbitrary friction; it is engineered conversion.

The \$99 Pro price point represents the highest amount a professional can spend without seeking approval from anyone — it falls below the typical corporate expense reporting threshold and is comparable to other SaaS subscriptions in a developer's toolkit (GitHub, Cursor, various API subscriptions). Pro+ at \$199 is exactly 2x Pro — a psychologically clean ratio that feels fair and easy to evaluate. Enterprise has no published price because funds of \$100M and \$5B AUM are anchored to completely different reference points for what software costs; a published price would undercut the larger segment. Ultra Secure at \$50K+ is deliberately high — it is a client filter. If the price causes hesitation, that client does not have alpha worth protecting.

Tier	Price	Credits/mo	Data	Simulations	Key Differentiator
Free	\$0	50	Yahoo 5yr	20 basic	Discovery, conversion funnel
Pro	\$99/mo (\$990/yr)	2,000	Polygon 20yr	100	Unlimited backtests, live trading
Pro+	\$199/mo (\$1,990/yr)	10,000	Polygon + L2	128	All models, unlimited teams
Enterprise	\$5K–\$15K/mo	100K+	Bloomberg+	128	Audit trail, SSO, SLA 99.99%
Ultra Secure	\$50K+/mo	Unlimited	Bloomberg+	128	Zero analysis, dedicated infra

Unit Economics by Tier

Tier	CAC	LTV	LTV:CAC	Avg Retention	Payback
Free → Pro	\$20	\$1,980 (20 mo)	99:1	20 months	<2 months
Pro+	\$50	\$6,570 (33 mo)	131:1	33 months	<4 months
Enterprise	\$5,000	\$180,000 (24 mo)	36:1	24 months	<1 month
Ultra Secure	\$10,000	\$1,800,000 (36 mo)	180:1	36 months	<1 month

Marketplace Revenue

The strategy marketplace creates a third revenue stream independent of subscriptions. Strategy developers publish their work at a monthly license fee of their choosing (typically \$20-\$200/month). QuantMaster takes a 30% platform fee.

By Year 3, with 1,000 listed strategies averaging \$50/month and an average of 30 subscribers each, the marketplace generates \$1.5M/month (\$18M/year) in additional revenue. This also creates powerful retention: strategy sellers have strong incentive to maintain their QuantMaster Pro+ subscription (to keep their seller status), and strategy buyers have strong incentive to stay subscribed (to maintain access to the strategies they use).

10. FINANCIAL PROJECTIONS (5 YEARS)

Revenue Build-Up

Year	Free Users	Pro	Pro+	Enterprise	Ultra Sec	MRR	ARR
1 end	50K	1,000	100	5	2	\$157K	\$1.9M
2 end	200K	8,000	600	40	10	\$1.17M	\$14M
3 end	600K	40,000	2,000	150	25	\$4.84M	\$58M
5 end	2M	200,000	15,000	800	100	\$33M+	\$400M+

Cost Structure

Cost Category	Year 1	Year 2	Year 3	Notes
Team (salaries)	\$2.5M	\$8M	\$20M	3→15→60 headcount
Infrastructure	\$180K	\$1M	\$6M	AWS, data feeds, CDN
Data (Polygon, Bloomberg)	\$60K	\$600K	\$3M	Scales with users
AI/Claude API	\$120K	\$500K	\$2M	Scales with usage
Sales & Marketing	\$800K	\$4M	\$12M	CAC investment
G&A, Legal, Compliance	\$500K	\$2M	\$5M	SOC2, legal, HR
TOTAL OPEX	\$4.16M	\$16.1M	\$48M	

Path to Profitability

Year	SaaS Revenue	SaaS OPEX	SaaS EBITDA	HF Profit	Combined
1	\$1.9M	\$4.16M	-\$2.26M	\$2.6M	+\$0.34M (breakeven)
2	\$14M	\$16.1M	-\$2.1M	\$15.8M	+\$13.7M
3	\$58M	\$48M	+\$10M	\$89.5M	+\$99.5M
5	\$400M+	\$180M	+\$220M	\$359M	+\$579M

Key insight: The hedge fund makes the combined entity profitable from Month 12, even while the SaaS is still investing in growth. This eliminates the 'valley of death' that kills most SaaS companies between Series A and Series B.

11. GO-TO-MARKET STRATEGY

Phase 1: Community-Led Growth (Months 1-6)

The initial growth strategy requires zero paid advertising. The target early adopter community — algorithmic traders on Reddit, Twitter/X, and Discord — is highly concentrated, technically sophisticated, and viral within itself. A single well-received post on r/algotrading can drive thousands of signups in a day. The ProductHunt launch is the anchor event: timed for a Tuesday morning Pacific time, with 200 upvote pledges secured in advance from the founder's network. The launch post should demonstrate the core magic in a 60-second screen recording: describe a strategy in English, watch the AI write the code, run the backtest, see the Deflated Sharpe warn that the strategy is overfit, watch the AI fix it. That loop, shown visually, sells itself.

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| • ProductHunt launch (Day 1) — target #1 Product of the Day |
| • r/algotrading, r/quant, r/investing posts — 500K combined subscribers |
| • Twitter/X thread: 'We built the most honest backtester ever' (Deflated Sharpe story) |
| • Hacker News Show HN post — reaches engineers and early adopters |
| • Discord server launch — build community before product launches |
| • YouTube: 60-second demo video, then weekly deep-dive tutorials |
| • Early adopter pricing: first 500 users get \$49/month (permanent, evangelists) |
| • Affiliate program: 30% recurring commission for referrals (month 2) |

Phase 2: Content-Led Growth (Months 4-12)

Content marketing targets the long tail of search intent: 'how to backtest a moving average strategy,' 'what is the Deflated Sharpe Ratio,' 'Heston model explained,' 'pairs trading Python tutorial.' These are high-intent searches with relatively low competition. Each article or video naturally demonstrates QuantMaster as the tool of choice. The educational content also serves as the foundation for the platform's built-in education module — every piece of content written for SEO becomes a lesson inside the product.

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| • SEO blog: 2 technical posts per week (educational, code-heavy) |
| • YouTube channel: 1 tutorial video per week (growing to 100K subscribers) |
| • Guest posts on Towards Data Science, Quantopian, Medium |
| • Twitter/X daily posts showing interesting backtest results (viral potential) |
| • Discord community grows to 10K members — becomes primary support channel |
| • Partnerships with quant finance courses (mention QuantMaster as tool) |

Phase 3: Enterprise Sales (Month 7 onwards)

Enterprise sales in quantitative finance follow the 'bottom-up' model pioneered by Slack and Figma: individual users adopt the tool on Pro tier, demonstrate value to their team, and then the organization upgrades to Enterprise. The sales cycle is therefore partially pre-validated. A hedge fund VP of Quant Research who has used QuantMaster personally

for three months is far easier to convert than a cold call. The VP Sales hire (Month 7) focuses exclusively on these warm inbound leads from individual users who want to bring the product to their firm.

12. TECHNICAL STACK & MVP TIMELINE

Architecture Philosophy

The technical architecture is designed for two constraints that seem contradictory: build an MVP in 8-10 weeks with 2-3 engineers, and build infrastructure that can scale to one million users without a complete rewrite. These are reconciled by choosing components that are simple at small scale and scalable at large scale. FastAPI (Python) handles all API logic — simple to write, easy to test, and deployable on any infrastructure. PostgreSQL with TimescaleDB extension handles all time-series storage — TimescaleDB has been shown to handle 100M+ rows with sub-second query performance. Redis handles all caching and real-time pub/sub. The backtester core is written in Rust (for speed) with Python bindings (for developer productivity). The AI agent calls the Claude API directly. Charts use Lightweight Charts (open source). Nothing proprietary or hard to replace.

Component	Technology	Why	Scaling Path
Frontend	React 18 + TypeScript + Vite	Fast dev, large talent pool	Next.js for SSR later
Charts	Lightweight Charts (open source)	TradingView quality, free	Same library at 10M users
Code Editor	Monaco Editor	VS Code engine, free	Same at any scale
Backend API	FastAPI (Python)	Async, easy, math-friendly	Microservices later
Database	PostgreSQL + TimescaleDB	Time-series optimized, SQL	Sharding at 10M+ rows
Cache	Redis	Sub-millisecond pub/sub	Redis Cluster at scale
Backtester	Rust core + Python bindings	2 years in 2 seconds	Distributed at scale
AI Agent	Claude API (Anthropic)	Best code generation	Fine-tune later
Deploy	AWS + GitHub Actions	Standard, fast setup	Multi-region later
Data	Yahoo + FRED + CoinGecko (free tier)	\$0 to start	+ Polygon, Bloomberg

MVP in 10 Weeks

Week	Milestone	Team
1-2	Core infrastructure: FastAPI, PostgreSQL, Redis, CI/CD, auth	2 engineers
3-4	Charts terminal (Lightweight Charts) + data feed (Yahoo + FRED)	2 engineers
5-6	Backtester (Backtrader + Rust slippage + Deflated Sharpe)	2 engineers
7-8	Monaco IDE + AI Agent integration + 20 math models	2 engineers + founder
9	Visual strategy builder + paper trading (Alpaca)	2 engineers + founder

10

Polish, QA, landing page, ProductHunt prep,
beta users

Full team

13. COMPETITIVE ANALYSIS

The Landscape

Three players dominate quantitative trading tooling today, each with significant market share and each with a clearly exploitable weakness. TradingView is the chart and discovery layer — 50 million users, beautiful interface, real-time data from hundreds of exchanges. Its weakness is depth: the backtester is toy-grade, the API is limited, and there is zero AI capability. QuantConnect is the execution layer for serious quants — event-driven backtesting, algorithm deployment, live trading. Its weakness is accessibility: the learning curve is extremely steep, the interface is developer-only, and there is no visual layer. Bloomberg is the data and analytics layer — comprehensive, trusted, institutional. Its weakness is cost (\$2,400/month per terminal), age (UI from 1990), and zero integration with modern AI workflows.

Feature	TradingView	QuantConnect	Bloomberg	QuantMaster
Charts	■■■■■	■■	■■	■■■■■
Backtester	■■	■■■■■	■■■■■	■■■■■
Math Models	■■	■■■	■■■■■	■■■■■
AI Generation	None	None	None	Full
Live Monitoring	None	None	None	Full
Simulations Library	None	Basic	Advanced	128 models
Team / Collaboration	Limited	Limited	Yes	Full
Price	\$15–\$200	\$150–\$250	\$2,400+	\$0–\$99
Overfitting Detection	None	Manual	Manual	Automatic

The core competitive positioning is: 'Bloomberg gives you data. QuantConnect lets you backtest. QuantMaster is the only platform that tells you when your strategy is too good to be true — before you put it in production and look bad in front of your CIO.' This framing is designed to resonate specifically with institutional users who are risk-conscious and whose primary fear is deploying a strategy that fails publicly.

14. LEGAL STRUCTURE & COMPLIANCE

Two Entities, Clean Separation

QuantMaster Inc. is incorporated as a C-Corporation in Delaware — the standard choice for VC-backed startups, providing the cleanest structure for institutional investment. The Terms of Service include an explicit, clearly-written clause: 'By using QuantMaster, you grant QuantMaster Inc. a perpetual, irrevocable, worldwide license to use anonymized, aggregated derivative data from your strategies for platform improvement, research, and analytics purposes. No individual strategy, code, or personally identifiable information will be shared with any third party.' This language is reviewed by privacy counsel and designed to withstand GDPR and CCPA scrutiny.

QuantMaster Capital LP is structured as a Delaware Limited Partnership. The general partner is a separate LLC owned by the founder. The fund files as a Registered Investment Adviser (RIA) with the SEC upon reaching \$25M AUM (required threshold). Below that threshold, it operates under the private fund exemption with a maximum of 15 sophisticated investors. The fund has a completely separate board, separate legal counsel, and a separate compliance officer.

15. TEAM STRUCTURE & HIRING

Phase	Headcount	Critical Hires	Monthly Burn
MVP (Weeks 1-10)	3	Founder + 2 Sr. Engineers	\$50K
Launch (Months 1-6)	8	+ VP Eng, Designer, PM, Marketer, 1 Eng	\$250K
Series A (Month 7-12)	20	+ VP Sales, 5 Eng, CSM, 2 Ops	\$700K
Year 2	50	+ CFO, Head of Data, 25 more	\$2M
Year 3	150	+ CIO (HF), compliance, 100 more	\$5M+

The most critical early hires are both senior engineers with quantitative finance experience — people who understand the mathematical correctness requirements of a backtester and who can implement Almgren-Chriss slippage or Deflated Sharpe from first principles. These are not typical software engineers; compensation must be at the top of market (\$200-300K+ cash in Year 1) or supplemented with meaningful equity.

16. RISK MANAGEMENT & MITIGATION

Risk	Probability	Impact	Mitigation
Data vendor price increases	Medium	High	Multi-vendor contracts, own data pipeline for basics
AI quality failures (bad code)	Medium	Medium	Automated validation pipeline, human review for Enterprise
Backtest-to-live divergence	High	High	Conservative cost assumptions, daily audit comparison
High user churn (>5% monthly)	Medium	High	Deep engagement features, marketplace lock-in
Slow Enterprise sales cycle	High	Medium	Bottom-up adoption (individual → team → fund)
Hedge fund underperformance	Medium	Low for SaaS	Separate entities, SaaS is independent
Competitor copies features	High	Medium	Execution speed, network effects, data moat
Data privacy regulation	Low	High	GDPR-grade from Day 1, ToS reviewed by counsel

17. EXIT STRATEGY & OPTIONALITY

Three Paths to Extraordinary Outcomes

The dual-entity structure creates exit optionality that is unusual in SaaS. Most SaaS companies have one exit: sell the company or IPO. QuantMaster has three independent paths, each of which generates extraordinary outcomes.

Option 1: Sell SaaS, Keep Hedge Fund (Recommended)

At Year 3-4, with \$56M+ ARR growing 80%+ and the platform's AI and simulation moat clearly established, the SaaS company is worth 20-30x ARR — a valuation of \$1.1B to \$1.7B. Strategic acquirers include Bloomberg (wants AI and retail distribution), Intercontinental Exchange (wants platform capabilities), Refinitiv/LSEG (wants quant tools), and Citadel Securities (wants data moat). After tax, the founder nets \$700M–\$1.1B in cash. The hedge fund continues generating \$30M–\$89M/year in perpetuity. Total 10-year outcome: \$1B+ in cash + \$300M+ in cumulative hedge fund distributions.

Option 2: IPO the SaaS Company

At Year 3-4 with \$56M ARR and clear path to \$400M, the IPO valuation is \$2.8B–\$3.5B (50-60x forward ARR, consistent with high-growth financial SaaS comps). The founder retains 25-30% post-dilution equity = \$700M–\$1B in public stock. The hedge fund remains private, unlisted, and continues compounding.

Option 3: Compound Both Indefinitely

No sale, no IPO. Run both businesses at maximum efficiency, reinvest profits, and compound over 20 years. By Year 10, the hedge fund at \$10B AUM generates \$400M+/year in net profit. The SaaS at \$1B+ ARR generates \$700M+/year in EBITDA. Combined: \$1.1B/year in cash generation. This is the path to building the next Citadel or Renaissance Technologies — private, profitable, generational.

CONCLUSION

QuantMaster is not a feature — it is a category. The quantitative trading software market has not seen a genuinely new approach in twenty years. Bloomberg is still the terminal from 1990. TradingView's backtester is still a toy. QuantConnect is still too hard for most traders. Nobody has applied AI to the full workflow. Nobody has built a simulation library at this depth. Nobody has created honest overfitting detection. Nobody has a 600,000-user data moat feeding a hedge fund. QuantMaster addresses all of these simultaneously — not because it is trying to do too many things, but because they are all parts of the same workflow that has never been unified before.

The financial case is clear. The technical case is clear. The market need is clear. The moat is real and defensible. The exit options are extraordinary. The only variable is execution — and execution starts with building the MVP.

Build it. ■